

Unlocking Customer Insights

A Comprehensive Retail Performance Analysis using Python,
PostgreSQL, and Power BI Visualization

THE PROJECT LIFECYCLE

An Integrated Workflow

This project demonstrates an end-to-end analytical pipeline. By combining robust data cleaning with complex relational querying and interactive visualization, we transform raw transactions into strategic business intelligence.

 **Python:** Advanced data cleaning and median imputation.

 **SQL:** Deep-dive behavioral analysis in PostgreSQL.

 **Power BI:** Executive performance dashboards.

Governance Pyramid PowerPoint Template

ance Pyramid: Clear Roles for Secure, Trusted Data



THE TECHNICAL STACK



Python

Used for feature engineering and handling missing ratings using median-per-category logic to maintain data integrity.



PostgreSQL

Implemented complex CTEs, Window Functions, and CASE statements to segment 3.9k+ customers into loyalty buckets.



Power BI

Designed a star-schema model with DAX Time Intelligence to track YTD growth and behavioral shifts.

The SQL Analytical Engine

Turning Relational Data into Strategic Gold

PRIMARY SQL AGGREGATIONS

Analysis Category	Key Metric	Statistical Result
Gender Revenue	Total Spend (Male)	\$157,890
Gender Revenue	Total Spend (Female)	\$75,191
Shipping Impact	Express Avg. Spend	\$60.48
Shipping Impact	Standard Avg. Spend	\$58.46
Loyalty Segments	Loyal Customer Count	3,116

MAJOR REVENUE DRIVER

\$157k

Male Customer Revenue

Target Demographic Insight

Initial SQL analysis reveals that male customers contribute significantly more to the total revenue than female customers. This insight allows for gender-specific marketing budget reallocation to maximize Return on Ad Spend (ROAS).

ANALYTICAL LOGIC: SEGMENTATION

The "Why" Behind the Query

To understand the business health, we implemented a **Customer Segmentation** model. Using SQL **CASE** logic, we categorized users by purchase frequency:

```
Segment = { 'Loyal' if Purchases > 10  
            'Returning' if Purchases 2-10  
            'New' if Purchases = 1
```

Impact of Logic

This allows the business to see that while "Loyal" customers are the majority (3,116), their average spend is similar to "New" customers, suggesting a need for loyalty incentives to increase order value.

Visual Intelligence

Interpreting Patterns in the Power BI Dashboard

BEHAVIORAL DASHBOARD

Executive Summary

The first dashboard layer focuses on **Customer Behavior**. It serves as a diagnostic tool for high-level management to track total customer health, average spending, and demographic preferences.

By integrating slicers for Gender and Category, the user can instantly pivot from "Clothing" trends to "Footwear" margins.

Customer Segmentation

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Quality Seekers

- Focus on product excellence.
- Value premium features.
- Willing to pay more for superior quality.
- Expect reliable performance and durability.
- Read reviews before making decisions.



Loyal Customers

- Prefer familiar brands.
- Show strong repeat purchase behavior.
- Respond well to loyalty programs.
- Share positive experiences with others.
- Stay connected through brand updates.

REVENUE BY AGE GROUP



Young Adults lead revenue contribution, identifying them as the primary target for growth campaigns.

PLANT CO. ANALYSIS

Advanced Performance Analysis

This second layer focuses on **Year-to-Date (YTD)** performance. It utilizes advanced DAX to compare current results against the same period last year (PYTD).

The Waterfall chart helps pinpoint the "leakage" in profit, specifically identifying which regions or categories failed to meet the previous year's benchmarks.



STRATEGIC RECOMMENDATIONS

- ★ **Loyalty Value Gap:** Implement tiered rewards for the "Loyal" segment (3k+ users) to increase their average order value from \$59 to \$75.
- 🚚 **Shipping Optimization:** Since Express shipping correlates with higher spend (\$60 vs \$58), offer free Express shipping for orders over \$75 to drive higher basket sizes.
- 👥 **Age-Targeted Marketing:** Focus digital ad spend on the "Young Adult" and "Middle Aged" segments, as they comprise over 50% of total revenue.
- 📊 **Subscription Revamp:** Current data shows subscribers spend no more than non-subscribers. Revise subscription benefits to include exclusive "Flash Sales."

Questions & Discussion

Thank you for your attention. This project is a testament to the power of structured data analysis in driving business growth.

Data Analysis Portfolio | 2024

IMAGE SOURCES



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